```
create database if not exists f1_raw;
```

OK

Create tables for CSV files

Create circuits table

```
DROP TABLE IF EXISTS f1_raw.circuits;
CREATE TABLE IF NOT EXISTS f1_raw.circuits
(circuitId INT,
circuitRef STRING,
name STRING,
location STRING,
country STRING,
lat DOUBLE,
lng DOUBLE,
alt INT,
url STRING
)
USING csv
OPTIONS (path "/mnt/formula1dl1204/raw/circuits.csv", header true)
```

OK

```
SELECT * FROM f1_raw.circuits;
```

Table

	circuitId	circuitRef	name	location	country	lat	lng
1	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.96
2	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.73
3	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.510
4	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.2611
5	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.40
6	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.4205
7	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	-73.522
8	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.1636
9	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.0169
10	10	hockenheimring	Hockenheimring	Hockenheim	Germany	49.3278	8.5658
11	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.248
12	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.33166
13	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.9713
14	14	monza	Autodromo Nazionale di Monza	Monza	Italy	45.6156	9.2811

77 rows

Create races table

```
DROP TABLE IF EXISTS f1_raw.races;
CREATE TABLE IF NOT EXISTS f1_raw.races
(raceId INT,
year INT,
round INT,
circuitId INT,
name STRING,
date DATE,
time STRING,
url STRING)
USING csv
OPTIONS (path "/mnt/formula1dl1204/raw/races.csv", header true)
```

OK

```
SELECT * FROM f1_raw.races;
```

Table

raceId	year	round	circuitId	name	date	time	url
26	2008	9	9	British Grand Prix	2008-07-06	12:00:00	http://en.wikipedia.org
27	2008	10	10	German Grand Prix	2008-07-20	12:00:00	http://en.wikipedia.org
28	2008	11	11	Hungarian Grand Prix	2008-08-03	12:00:00	http://en.wikipedia.org
29	2008	12	12	European Grand Prix	2008-08-24	12:00:00	http://en.wikipedia.org
30	2008	13	13	Belgian Grand Prix	2008-09-07	12:00:00	http://en.wikipedia.org
31	2008	14	14	Italian Grand Prix	2008-09-14	12:00:00	http://en.wikipedia.org
32	2008	15	15	Singapore Grand Prix	2008-09-28	12:00:00	http://en.wikipedia.org
33	2008	16	16	Japanese Grand Prix	2008-10-12	04:30:00	http://en.wikipedia.org
34	2008	17	17	Chinese Grand Prix	2008-10-19	07:00:00	http://en.wikipedia.org
35	2008	18	18	Brazilian Grand Prix	2008-11-02	17:00:00	http://en.wikipedia.org
36	2007	1	1	Australian Grand Prix	2007-03-18	03:00:00	http://en.wikipedia.org
37	2007	2	2	Malaysian Grand Prix	2007-04-08	07:00:00	http://en.wikipedia.org
38	2007	3	3	Bahrain Grand Prix	2007-04-15	11:30:00	http://en.wikipedia.org
39	2007	4	4	Spanish Grand Prix	2007-05-13	12:00:00	http://en.wikipedia.org
40	2007	5	6	Monaco Grand Prix	2007-05-27	12:00:00	http://en.wikipedia.org
41	2007	6	7	Canadian Grand Prix	2007-06-10	17:00:00	http://en.wikipedia.org

1,058 rows

Create tables for JSON files

Create constructors table

- Single Line JSON
- Simple structure

```
DROP TABLE IF EXISTS f1_raw.constructors;
CREATE TABLE IF NOT EXISTS f1_raw.constructors
(
constructorId INT,
constructorRef STRING,
name STRING,
nationality STRING,
url STRING)
USING json
OPTIONS(path "/mnt/formula1dl1204/raw/constructors.json")
```

OK

```
SELECT * FROM f1_raw.constructors;
```

Table

	constructorId	constructorRef	name	nationality	url
1	1	mclaren	McLaren	British	http://en.wikipedia.org/wiki/McLaren
2	2	bmw_sauber	BMW Sauber	German	http://en.wikipedia.org/wiki/BMW_Sauber
3	3	williams	Williams	British	http://en.wikipedia.org/wiki/Williams_Grand_Prix_Engineering
4	4	renault	Renault	French	http://en.wikipedia.org/wiki/Renault_in_Formula_One
5	5	toro_rosso	Toro Rosso	Italian	http://en.wikipedia.org/wiki/Scuderia_Toro_Rosso
6	6	ferrari	Ferrari	Italian	http://en.wikipedia.org/wiki/Scuderia_Ferrari
7	7	toyota	Toyota	Japanese	http://en.wikipedia.org/wiki/Toyota_Racing
8	8	super_aguri	Super Aguri	Japanese	http://en.wikipedia.org/wiki/Super_Aguri_F1
9	9	red_bull	Red Bull	Austrian	http://en.wikipedia.org/wiki/Red_Bull_Racing
10	10	force_india	Force India	Indian	http://en.wikipedia.org/wiki/Racing_Point_Force_India
11	11	honda	Honda	Japanese	http://en.wikipedia.org/wiki/Honda_Racing_F1
12	12	spyker	Spyker	Dutch	http://en.wikipedia.org/wiki/Spyker_F1
13	13	mf1	MF1	Russian	http://en.wikipedia.org/wiki/Midland_F1_Racing
14	14	spyker_mf1	Spyker MF1	Dutch	http://en.wikipedia.org/wiki/Midland_F1_Racing

211 rows

Create drivers table

- Single Line JSON
- Complex structure

```
DROP TABLE IF EXISTS f1_raw.drivers;
CREATE TABLE IF NOT EXISTS f1_raw.drivers
(
driverId INT,
driverRef STRING,
number INT,
code STRING,
name STRUCT<forename: STRING, surname: STRING>,
dob DATE,
nationality STRING,
url STRING)
USING json
OPTIONS (path "/mnt/formula1dl1204/raw/drivers.json")
```

OK

Create results table

- Single Line JSON
- Simple structure

```
DROP TABLE IF EXISTS f1_raw.results;
CREATE TABLE IF NOT EXISTS f1_raw.results(
resultId INT,
raceId INT,
driverId INT,
constructorId INT,
number INT,grid INT,
position INT,
positionText STRING,
positionOrder INT,
points INT,
laps INT,
time STRING,
milliseconds INT,
fastestLap INT,
rank INT,
fastestLapTime STRING,
fastestLapSpeed FLOAT,
statusId STRING)
USING json
OPTIONS(path "/mnt/formula1dl1204/raw/results.json")
```

OK

```
SELECT * FROM f1_raw.results
```

Table

	resultId	raceId	driverId	constructorId	number	grid	position	positionTe
80	80	21	3	3	7	15	null	
81	81	21	4	4	5	2	null	R
82	82	21	22	11	17	11	null	R
83	83	21	5	1	23	6	null	R
84	84	21	19	8	19	21	null	R
85	85	21	7	5	14	16	null	R
86	86	21	12	4	6	10	null	R
87	87	21	16	10	20	20	null	R
88	88	21	20	5	15	18	null	R
89	89	22	13	6	2	1	1	1
90	90	22	1	1	22	3	2	2
91	91	22	8	6	1	4	3	3
92	92	22	9	2	4	5	4	4
93	93	22	2	2	3	9	5	5
94	94	22	4	4	5	7	6	6

10,000+ rows | Truncated data due to row limit

Create pit stops table

- Multi Line JSON
- Simple structure

```
DROP TABLE IF EXISTS f1_raw.pit_stops;
CREATE TABLE IF NOT EXISTS f1_raw.pit_stops(
driverId INT,
duration STRING,
lap INT,
milliseconds INT,
raceId INT,
stop INT,
time STRING)
USING json
OPTIONS(path "/mnt/formula1dl1204/raw/pit_stops.json", multiLine true)
```

OK

```
SELECT * FROM f1_raw.pit_stops;
```

Table

	driverId	duration	lap	milliseconds	raceId	stop	time
1	153	26.898	1	26898	841	1	17:05:23
2	30	25.021	1	25021	841	1	17:05:52
3	17	23.426	11	23426	841	1	17:20:48
4	4	23.251	12	23251	841	1	17:22:34
5	13	23.842	13	23842	841	1	17:24:10
6	22	23.643	13	23643	841	1	17:24:29
7	20	22.603	14	22603	841	1	17:25:17
8	814	24.863	14	24863	841	1	17:26:03
9	816	25.259	14	25259	841	1	17:26:50
10	67	25.342	15	25342	841	1	17:27:34
11	2	22.994	15	22994	841	1	17:27:41
12	1	23.227	16	23227	841	1	17:28:24
13	808	24.535	16	24535	841	1	17:28:39
14	3	23.716	16	23716	841	1	17:29:00
15	155	24.064	16	24064	841	1	17:29:06

8,030 rows

Create tables for list of files

Create Lap Times Table

- CSV file
- Multiple files

```
DROP TABLE IF EXISTS f1_raw.lap_times;
CREATE TABLE IF NOT EXISTS f1_raw.lap_times(
raceId INT,
driverId INT,
lap INT,
position INT,
time STRING,
milliseconds INT
)
USING csv
OPTIONS (path "/mnt/formula1dl1204/raw/lap_times")
```

OK

```
SELECT * FROM f1_raw.lap_times
```

Table

	raceId	driverId	lap	position	time	milliseconds
26	841	20	26	1	1:31.328	91328
27	841	20	27	1	1:31.568	91568
28	841	20	28	1	1:31.113	91113
29	841	20	29	1	1:31.339	91339
30	841	20	30	1	1:31.054	91054
31	841	20	31	1	1:31.707	91707
32	841	20	32	1	1:31.611	91611
33	841	20	33	1	1:31.406	91406
34	841	20	34	1	1:31.871	91871
35	841	20	35	1	1:32.597	92597
36	841	20	36	1	1:53.737	113737
37	841	20	37	1	1:39.321	99321
38	841	20	38	1	1:33.632	93632
39	841	20	39	1	1:31.005	91005
40	841	20	40	1	1:30.530	90530
41	841	20	41	1	1:30.140	90140
42	841	20	42	1	1:30.419	90419

10,000+ rows | Truncated data due to row limit

Create Qualifying Table

- JSON file
- MultiLine JSON
- Multiple files

```
DROP TABLE IF EXISTS f1_raw.qualifying;
CREATE TABLE IF NOT EXISTS f1_raw.qualifying(
constructorId INT,
driverId INT,
number INT,
position INT,
q1 STRING,
q2 STRING,
q3 STRING,
qualifyId INT,
raceId INT)
USING json
OPTIONS (path "/mnt/formula1dl1204/raw/qualifying", multiLine true)
```

OK

```
SELECT * FROM f1_raw.qualifying
```

Table

	constructorId	driverId	number	position	q1	q2	q3	qualifyId
1	1	1	22	1	1:26.572	1:25.187	1:26.714	1
2	2	9	4	2	1:26.103	1:25.315	1:26.869	2
3	1	5	23	3	1:25.664	1:25.452	1:27.079	3
4	6	13	2	4	1:25.994	1:25.691	1:27.178	4
5	2	2	3	5	1:25.960	1:25.518	1:27.236	5

6	7	15	11	6	1:26.427	1:26.101	1:28.527	6
7	3	3	7	7	1:26.295	1:26.059	1:28.687	7
8	9	14	9	8	1:26.381	1:26.063	1:29.041	8
9	7	10	12	9	1:26.919	1:26.164	1:29.593	9
10	5	20	15	10	1:26.702	1:25.842	\N	10
11	11	22	17	11	1:26.369	1:26.173	\N	11
12	4	4	5	12	1:26.907	1:26.188	\N	12
13	11	18	16	13	1:26.712	1:26.259	\N	13
14	3	6	8	14	1:26.891	1:26.413	\N	14

8,694 rows

Table

	col_name	data_type	comment
1	constructorId	int	null
2	driverId	int	null
3	number	int	null
4	position	int	null
5	q1	string	null
6	q2	string	null
7	q3	string	null
8	qualifyId	int	null
9	raceId	int	null
10			
11	# Detailed Table Information		
12	Catalog	spark_catalog	
13	Database	f1_raw	
14	Table	qualifying	
15	Owner	root	

25 rows